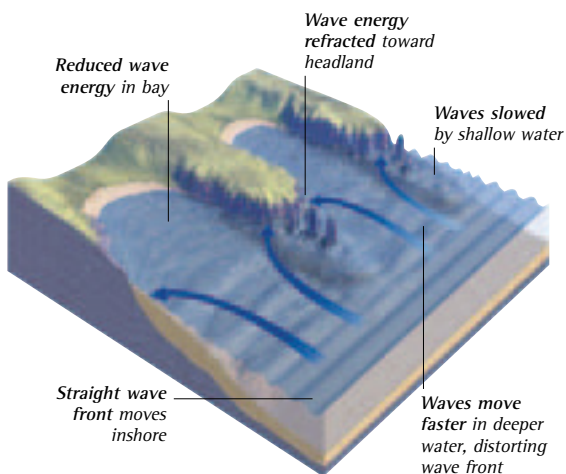




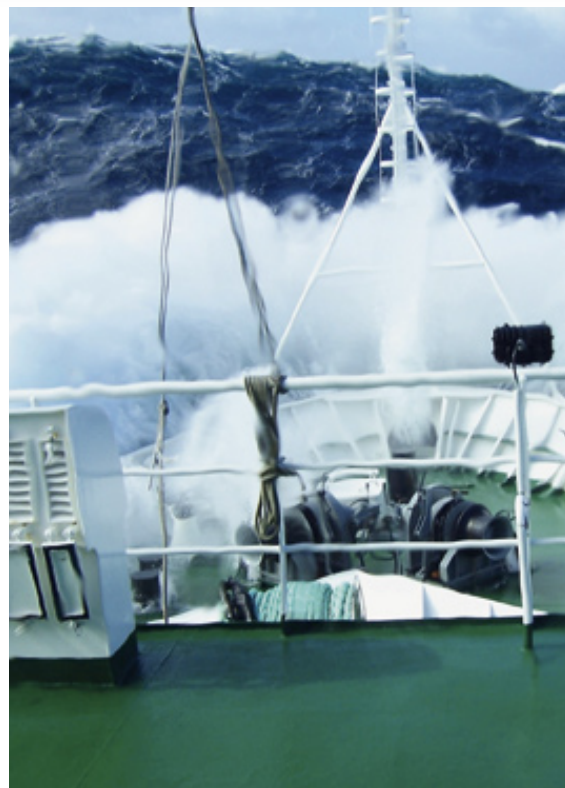
▲ PLUNGING BREAKERS

The farther a wave travels, the bigger it gets. The biggest waves occur in the Southern Ocean, where strong winds sweep them eastward around Antarctica. But some also head north into the Pacific. They may grow to be 60 ft (18 m) high by the time they reach the steep shores of Hawaii, which transform them into huge plunging breakers.



◀ WAVE REFRACTION

At times, waves reach some parts of the shore before others. The shallow patches slow them down, distorting the wave pattern so the straight wave crests become curved. This tends to divert the destructive wave energy away from sheltered bays, and focuses it on the exposed headlands.



▲ ROGUE WAVES

Out at sea, regular swells can be quite high without being particularly dangerous. But if two swells with different wavelengths come together, they form a more complex wave pattern. This has flat spots where the wave troughs of one swell coincide with the crests of another, but has extra-large waves where two wave crests reinforce each other, as seen here. Occasionally, colossal waves form where opposing storm waves meet. These may be as high as 100 ft (30 m), and they often have breaking crests. Such rogue waves have been known to sweep right over huge ships, causing immense damage. They are probably responsible for many ships being lost without a trace.

DESTRUCTIVE POWER

When a wave breaks, most of its mass topples forward with tremendous force. In a major storm, this force can reach up to 6,000 lb per sq ft (30,000 kg per sq m). On some shores much of this energy is absorbed by banks of shingle originally thrown up by the waves, so these act as natural breakwaters. On others, however, there may be nothing to stop the full force of the breaking waves slamming into coastal structures and cliffs. Over time, whole towns may disappear as the rock beneath them is swept away by the sea.

